

week-02-html

Week 02 — HTML Deep Dive

Session goal: by the end of class you have a valid, semantic profile page with a working form.

Resource: [HTML Online Editor \(Compiler, Interpreter & Runner\)](#).

HTML describes the structure of a web page. HTML elements tell the browser what each piece of content is: a heading, a paragraph, a link, a form, and so on.

What we covered (Week 01)

- Client vs. server
 - Browser and server communication
 - HTTP requests and responses
 - URL anatomy
 - HTTP methods
 - HTTP status codes
 - HTML, CSS, and JavaScript roles
 - Development environment setup
 - Node.js, npm, and Git basics
 - Extensions: Live Server + Prettier
 - Creating and serving an HTML page
 - Basic HTML document structure
 - DevTools: Elements, Console, and Network
 - Reading Network requests and status codes
 - Basic debugging habits
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Warm-up (5 min)

Open last week's `hello-web/index.html`. Add:

1. A second paragraph about yourself
2. An unordered list of three skills
3. An image with an `alt` attribute

Result

Hello, Web Engineering

My name is **Your Name** and this is my first page.

I enjoy learning web development and building interactive websites.

- HTML
- CSS
- JavaScript



Click me

A Simple HTML Document

Every HTML page needs the same basic skeleton.

Example

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8"/>
    <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
    <meta name="description" content="A short page summary for search engines."/>
    <title>My Profile</title>
  </head>
  <body>
    <h1>Hello</h1>
    <p>This is my first paragraph.</p>
  </body>
</html>
```

Example Explained

Part	Meaning
<code><!DOCTYPE html></code>	This document is HTML5
<code><html></code>	The root element of the page

Part	Meaning
<code><head></code>	Information about the page (not shown on screen)
<code><title></code>	Text in the browser tab
<code><body></code>	All visible content
<code><h1></code>	A large heading
<code><p></code>	A paragraph

Note: If you forget `charset="UTF-8"`, special characters can look broken. If you forget `viewport`, the page looks tiny on phones.

Class activity A — blank file challenge (8 min)

1. Create folder `week-02` and file `profile.html`
2. Type the skeleton from memory (doctype, html, head, body)
3. Open it with Live Server

Done when: the browser tab shows your title.

HTML Elements

An HTML element has a start tag, content, and an end tag:

```
<tagname>Content goes here...</tagname>
```

Example

```
<h1>My First Heading</h1>
<p>My first paragraph.</p>
<a href="https://example.com">Visit Example</a>
```

Example Explained

Start tag	Content	End tag
<code><h1></code>	My First Heading	<code></h1></code>
<code><p></code>	My first paragraph.	<code></p></code>
<code><a></code>	Visit Example	<code></code>

Note: Some elements have no content and no end tag. These are called **empty elements** (void elements):

`<img>`, `<br>`, `<hr>`, `<input>`, `<meta>`, `<link>`.

Example — Attributes

Attributes give extra information about an element. They go in the start tag:

```
<a href="https://example.com" target="_blank">Visit site</a>
```

- `href` = where the link goes

- `target="_blank"` = open in a new tab

HTML Headings

HTML headings are defined with `<h1>` to `<h6>`.

`<h1>` is the most important. `<h6>` is the least important.

Example

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
```

Example Explained

- Use **one** `<h1>` per page (the page title)
- Do not skip levels (do not jump from `h1` to `h3`)
- Choose headings for structure, not for font size (CSS controls size later)

HTML Paragraphs and Text

Example

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

<p>
  <strong>Important text</strong> and
  <em>emphasized text</em>.
</p>
```

Example Explained

Tag	Use
<code><p></code>	A paragraph
<code></code>	Important text (usually bold)
<code></code>	Emphasized text (usually italic)

HTML Lists

Example — Unordered List

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
```

```
<li>Milk</li>
</ul>
```

Example — Ordered List

```
<ol>
  <li>Open the editor</li>
  <li>Create a file</li>
  <li>Save the file</li>
</ol>
```

Example — Description List

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

Example Explained

Tag	Use
	Bullet list (order does not matter)
	Numbered list (order matters)
	One list item
<dl> / <dt> / <dd>	Term and definition

HTML Links

HTML links are defined with the `<a>` tag. The `href` attribute is the destination.

Example

```
<!-- Link to another page -->
<a href="about.html">About us</a>

<!-- Link to another website -->
<a href="https://www.w3schools.com" target="_blank" rel="noopener noreferrer">
  Visit W3Schools
</a>

<!-- Jump to a section on the same page -->
<a href="#contact">Contact</a>

<!-- Email and phone -->
```

```
<a href="mailto:me@example.com">Email me</a>
<a href="tel:+923001234567">Call me</a>
```

Example Explained

Attribute / pattern	Meaning
<code>href="page.html"</code>	Link to another file
<code>href="https://..."</code>	Link to an external site
<code>href="#id"</code>	Jump to an element with that <code>id</code>
<code>target="_blank"</code>	Open in a new tab
<code>rel="noopener noreferrer"</code>	Safer new-tab links (always use with <code>target="_blank"</code>) <code>noopener</code> (Security Focus) and <code>noreferrer</code> (Privacy Focus): prevents a newly opened page from hacking the original tab (<code>noopener</code>) and hides the originating website's URL from analytics tracking (<code>noreferrer</code>) SEO Impact: It has zero effect on search rankings or link equity. Search engines like Google crawl and pass link authority through these tags normally

Example — Link Target

```
<section id="contact">
  <h2>Contact</h2>
  <p>Write to me below.</p>
</section>
```

The link `<a href="#contact">Contact</a>` scrolls to that section.

HTML Images

HTML images are defined with the `<img>` tag. It is an empty element (no end tag).

Example

```

```

Example Explained

Attribute	Meaning
<code>src</code>	Path or URL of the image

Attribute	Meaning
<code>alt</code>	Text if the image cannot load / for screen readers (required)
<code>width</code> / <code>height</code>	Size in pixels (helps stop page jump while loading)
<code>loading="lazy"</code>	Load the image only when it is near the screen

Tip: Write `alt` that describes the image. If the image is only decoration, use `alt=""`.

Example — Image with Caption

```
<figure>
  
  <figcaption>My Week 01 hello page on Live Server</figcaption>
</figure>
```

HTML Semantic Elements

Semantic elements clearly describe their meaning.

Example — Bad (no meaning)

```
<div class="header">...</div>
<div class="nav">...</div>
<div class="content">...</div>
<div class="footer">...</div>
```

Example — Good (semantic)

```
<body>
  <header>
    <h1>My Site</h1>
    <nav>
      <ul>
        <li><a href="profile.html">Home</a></li>
        <li><a href="contact.html">Contact</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <article>
      <h2>About me</h2>
```

```

<p>I am learning web engineering.</p>

<section id="skills">
  <h3>Skills</h3>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
  </ul>
</section>
</article>

<aside>
  <h2>Currently learning</h2>
  <p>JavaScript</p>
</aside>
</main>

<footer>
  <p>&copy; 2026 My Site</p>
</footer>
</body>

```

Example Explained

Tag	Meaning
<header>	Introductory content
<nav>	Navigation links
<main>	Main content (one per page)
<article>	Content that makes sense alone
<section>	A thematic group (usually with a heading)
<aside>	Side content (related links, tips)
<footer>	Footer for the page or section
<div>	Use only when no better tag exists

Class activity B — build the bones (12 min)

In `profile.html`, build:

1. `<header>` with your name (`<h1>`) and `<nav>` (Home / About / Contact)
2. `<main>` → `<article>` with a short bio
3. `<section>` with 4 skills in a `<ul>`
4. `<aside>` with “Currently learning...”
5. `<footer>` with copyright + `mailto:` link

Done when: View Source shows semantic tags, not only `<div>`.

Try it — Links and Images

Class activity C — media + links (10 min)

On your profile page:

1. Add one `<figure>` with `<figcaption>` and a real `alt`
2. Add three external links in the footer (all with `rel="noopener noreferrer"`)
3. Add a nav link `href="#skills"` and put `id="skills"` on your skills list

Done when: the skills link scrolls to the list.

HTML Forms

An HTML form collects user input. The `<form>` element wraps the inputs.

Example

```
<form action="#" method="post">
  <label for="name">Full name</label>
  <input type="text" id="name" name="name" required/>

  <label for="email">Email</label>
  <input type="email" id="email" name="email" required/>

  <button type="submit">Send</button>
</form>
```

Example Explained

Part	Meaning
<code><form></code>	Container for inputs
<code>action</code>	Where to send the data (server URL)
<code>method</code>	Usually <code>get</code> or <code>post</code>
<code><label for="id"></code>	Label linked to input with matching <code>id</code>
<code>name</code>	Name of the value when the form is submitted
<code>required</code>	Browser blocks empty submit

Important: Every input needs a `<label>` and a `name` .

Example — Input Types

```
<input type="text"/>
<input type="email"/>
<input type="password"/>
<input type="tel"/>
<input type="number"/>
<input type="date"/>
```

```

<input type="url"/>
<input type="file"/>
<input type="checkbox"/>
<input type="radio"/>
<input type="submit"/>

```

type	What you get
text	Normal text field
email	Checks email format + email keyboard on phones
password	Hidden characters
tel	Phone keypad on phones
number	Number field
date	Date picker
checkbox	On/off option
radio	Choose one option from a group

Example — Full Contact Form

```

<form action="#" method="post">
  <div>
    <label for="name">Full name</label>
    <input type="text" id="name" name="name" required autocomplete="name" />
  </div>

  <div>
    <label for="email">Email</label>
    <input
      type="email"
      id="email"
      name="email"
      required
      placeholder="you@example.com"
    />
  </div>

  <div>
    <label for="phone">Phone</label>
    <input type="tel" id="phone" name="phone" />
  </div>

  <fieldset>
    <legend>Preferred track</legend>
    <label>
      <input type="radio" name="track" value="frontend" /> Frontend
    </label>
    <label>

```

```

        <input type="radio" name="track" value="backend" /> Backend
    </label>
</fieldset>

<div>
    <label for="topic">Topic</label>
    <select id="topic" name="topic">
        <option value="">Choose one</option>
        <option value="help">Help</option>
        <option value="feedback">Feedback</option>
    </select>
</div>

<div>
    <label for="message">Message</label>
    <textarea id="message" name="message" rows="4"></textarea>
</div>

<label>
    <input type="checkbox" name="terms" required/>
    I accept the terms
</label>

<button type="submit">Send</button>
</form>

```

Example Explained

- `<textarea>` = multi-line text
- `<select>` + `<option>` = dropdown
- `<fieldset>` + `<legend>` = group related controls
- `placeholder` is a hint only — **not** a label

Class activity D — contact form (15 min)

Add a Contact section with a form with all the below inp:

- `text`, `email`, `tel`
- `select`
- `textarea`
- `checkbox`
- submit button

Then test:

1. Remove one `for` / `id` pair → label click should fail
2. Put it back

3. Submit empty → browser should block because of `required`

Done when: every label focuses its field.

HTML Tables

Use tables for **data**, never for page layout.

Example

```
<table>
  <caption>My tools</caption>
  <thead>
    <tr>
      <th scope="col">Tool</th>
      <th scope="col">Comfort</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">HTML</th>
      <td>Strong</td>
    </tr>
    <tr>
      <th scope="row">CSS</th>
      <td>Beginner</td>
    </tr>
    <tr>
      <th scope="row">Git</th>
      <td>Comfortable</td>
    </tr>
  </tbody>
</table>
```

Example Explained

Tag	Meaning
<code><table></code>	The table
<code><caption></code>	Table title
<code><thead></code> / <code><tbody></code>	Header and body sections
<code><tr></code>	Table row
<code><th></code>	Header cell
<code><td></code>	Data cell
<code>scope="col"</code> / <code>scope="row"</code>	What the header describes

Class activity E — mini data table (5 min)

Add a small table of 3 tools/courses and your comfort level. Include `<caption>`, `<thead>`, and `scope`.

HTML `id` and `class`

The `id` and `class` attributes name elements so you can link to them, style them (Week 03), or find them with JavaScript later.

Example

```
<h2 id="skills">My Skills</h2>
<p class="bio">I am learning web engineering.</p>
<p class="bio highlight">HTML is my favorite so far.</p>

<a href="#skills">Jump to skills</a>
```

Example Explained

Attribute	Rule	Use
<code>id</code>	Must be unique on the page	One specific element (<code>#skills</code>)
<code>class</code>	Can be reused on many elements	A group of similar elements (<code>.bio</code>)
Many classes	Separate with a space	<code>class="bio highlight"</code>

Note: `id` names should not start with a number. Prefer lowercase and hyphens: `contact-form`, not `Contact Form`.

Example — Comments

HTML comments are not shown in the browser. Use them as notes for yourself.

```
<!-- This is a comment -->
<!-- Main profile content starts here -->
<article>
  <h2>About me</h2>
</article>
```

Example Explained

- Start with `<!--`
- End with `-->`
- Comments help you mark sections while you build
- Do not put secret passwords in comments — anyone can View Source

Class activity F — name your sections (8 min)

On `profile.html`:

1. Give these unique `id` s: `skills`, `contact`, `about`
2. Add a nav link to each (`href="#skills"`, etc.) and click to confirm they jump

3. Add `class="card"` to your `<article>` and `class="muted"` to one paragraph (classes do nothing yet — that is fine)
4. Add three HTML comments that label: header, main content, form

Done when: clicking each nav link scrolls to the right section, and View Source shows your comments.

Lab (in class) — finish and validate

Finish `profile.html` :

- ☐ Full document skeleton (`<!DOCTYPE>` , `lang` , charset, viewport, description, title)
- ☐ Semantic tags: `header` , `nav` , `main` , `article` , `section` , `aside` , `footer`
- ☐ At least one `<figure>` + `<figcaption>` with good `alt`
- ☐ Form with **5+** input types, each with `label` + `name`
- ☐ Useful `id` s for in-page links + at least two `class` names ready for CSS
- ☐ Zero errors at validator.w3.org

Pass criteria: validator is green, and clicking a label focuses its input.

Submit: `profile.html` (+ images) and a screenshot of the validator result.

Homework

Due before Week 03.

Task 1 — Semantic teardown

Screenshot any real homepage. Draw boxes and label: `header` , `nav` , `main` , `article` , `aside` , `footer` .

Submit: annotated image.

Task 2 — Fix bad HTML

Rewrite this in `homework-rewrite.html` using semantic tags:

```
<div class="top">
  <div class="big-text">My Blog</div>
  <div class="links"><span onclick="go( '/' )">Home</span></div>
</div>
<div class="post">
  <div class="post-title">First Post</div>
  <div class="post-body">Hello world</div>
  <div class="date">Aug 9 2026</div>
</div>
```

Submit: file that passes the W3C validator.

Task 3 — Second page

1. Create `contact.html` with a form

2. Link `profile.html` ↔ `contact.html` in both navs
3. Same copyright line in both footers

Submit: both HTML files.

Task 4 — Reflect (max 5 sentences)

1. Where you almost used a `<div>` and what you used instead

Common Mistakes

Mistake	Problem	Fix
Only <code><div></code> tags	No meaning for browser / screen readers	Use semantic tags
Missing <code>alt</code>	Image has no text alternative	Add a real description
Label not linked	Clicking label does nothing	<code>for</code> must match <code>id</code>
Skip heading levels	Broken page outline	<code>h1</code> then <code>h2</code> then <code>h3</code>
Use <code>
</code> for space	Hard to maintain	Use CSS in Week 03
<code>target="_blank"</code> alone	Less safe	Add <code>rel="noopener noreferrer"</code>

Checklist before Week 03

- ☐ I can write a basic HTML document
- ☐ I know when to use `article`, `section`, `aside`, `div`
- ☐ Every image has `alt`
- ☐ Every input has a `label` and a `name`
- ☐ I know the difference between `id` and `class`
- ☐ My page passes the W3C validator